

SEC Daily Update System

Feasibility and MVP Design Report

Version date: 2026-07-01 · Purpose: explain what this round of validation did, how it was verified, what conclusions were reached, and how the daily update system should be designed.

This validation round has proven that the following route works end to end, and that it should be productized as a **filing acquisition system**: every day, first discover all newly accepted filings across the entire SEC market, then follow each accession to retrieve its materials, identify party roles, classify the documents, and extract 8-K events and XBRL financial facts. The route map below is also the structure of this report.

Color code used throughout: ■ measured & verified this round ■ gap / trap / caveat ■ design decision

THE VALIDATED ROUTE, STAGE BY STAGE

1	Daily index Discovery layer — what filed today, market-wide	\$3-\$4
2	Accession materials Everything behind one accession number	\$4-\$5
3	SGML header "Who + what happened" metadata layer	\$5-\$7
4	Document manifest Materials layer — which files, routed to which parser	\$7
5	Primary document Entry point into the report body · 8-K items = event-classification layer	\$8-\$9
6	XBRL / iXBRL instance Accession-level source of financial facts	\$10-\$11
7	Fact extraction Contexts, units, concepts, values → classified candidates	\$11-\$13
8	companyfacts crosscheck SEC company-level aggregation-and-reconciliation layer · weekly full index = PAC correction layer	\$12, \$14-\$17

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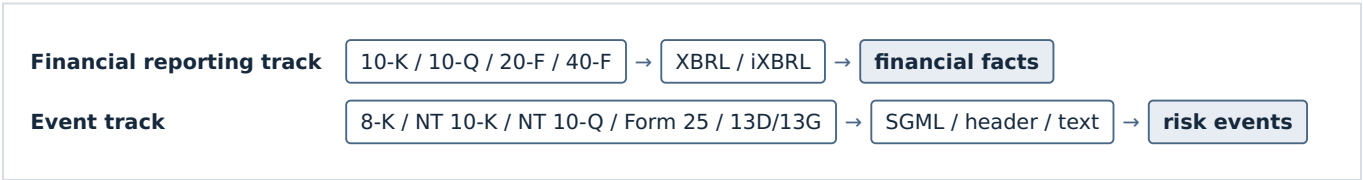
1

Executive Summary

This validation round has proven that the following route works end to end: **daily index → accession materials → SGML header → document manifest → primary document → XBRL/iXBRL instance → fact extraction → companyfacts crosscheck**.

This route should be productized as a **filing acquisition system**: every day, first discover all newly accepted filings across the entire SEC market, then follow each accession to retrieve its materials, identify party roles, classify the documents, and extract 8-K events and XBRL financial facts.

The value of the route is that two tracks hold simultaneously:



CAVEAT — THE ONE FAILURE MODE OBSERVED

Of the 17 financial-report samples in which an XBRL instance was detected this round, 16 yielded usable facts, and the remaining 1 was classified as **XBRL_EXISTS_BUT_NO_USEFUL_FACTS**. This failure is neither a download failure nor an XML parsing failure. It means: "an XBRL/iXBRL instance was detected, and the XML parsed successfully, but none of the useful financial candidates we defined — e.g. cash, assets, liabilities, revenue, net income — were extracted."

2

What Was Done

This round validated the critical path of the production system; it did not attempt to build the full product. The main work completed:

1.

Downloaded and parsed the daily master index for the fixed date 2026-06-26 .
2.

Downloaded and parsed the daily master index for the most recent valid EDGAR business day.
3.

Parsed the five daily-index fields: CIK, company name, form type, filed date, and file name.
4.

Derived from the file name: the accession number, the no-dash accession number, the complete-submission text URL, the .hdr.sgml URL, the directory index.json URL, and the filing-detail URL.

- 5. Sampled 105 accessions, covering 8-K, Form 4, 10-K/10-Q, 6-K, 13D/13G, registration/prospectus-type filings, and more.
- 6. For every sample, attempted to fetch `.hdr.sgml`, a capped read of the complete text, the directory `index.json`, the filing detail page, and the primary document.
- 7. Parsed the SGML header, extracting FILER, ISSUER, REPORTING OWNER, SUBJECT COMPANY, FILED BY, ITEM INFORMATION, and related metadata.
- 8. Ran item extraction on 20 8-K samples, achieving a 100% extraction rate.
- 9. Detected XBRL / iXBRL instances on the financial-report samples and actually parsed facts from them.
- 10. Ran a companyfacts crosscheck on the accessions whose XBRL was parsed.
- 11. Ran golden assertions — e.g. the row count of the fixed-date daily index, Apple's first-XBRL dates, the PCS Edventures 10-K fact example, the TSM `isXBRL` vs. companyfacts gap, and related checks.

Key measured results

105 Sampled accessions 8-K, Form 4, 10-K/10-Q, 6-K, 13D/13G, registration/prospectus types, and more.	100% SGML header identified Every sample yielded <code>.hdr.sgml</code> or a Range-capped header read of the complete <code>.txt</code>; the parser extracted at least the accession-level header facts or a party block.
100% index.json identified The accession-directory <code>index.json</code> URL derived from the daily-index <code>file_name</code> returned HTTP 200.	100% Primary document identified Every accession produced a non-empty <code>primary_document_name</code> from the merged document list.
20/20 8-K item extraction At least one standard 8-K item code was extracted from the accession header.	16 / 17 XBRL instances → usable facts Usable facts = cash / assets / liabilities / revenue / net_income.

Financial-report funnel	35 samples (10-K, 10-K/A, 10-Q, 10-Q/A, 20-F, 20-F/A, 40-F, 40-F/A, 6-K, 6-K/A) →
	17 XBRL instance detected → 16 usable facts
	+ 1 XBRL_EXISTS_BUT_NO_USEFUL_FACTS

3 What the Daily Index Is, and What a Row Looks Like

The **daily index** is the market-wide filing index file that SEC publishes on every EDGAR business day. It is the discovery layer, not the complete-materials layer. Per SEC's official documentation, each index record contains five fields: company name, form type, CIK, date filed, and file name. The file name includes the folder path, so it can be used to locate the complete submission text.

SAMPLE FILE HEADER

CIK	Company Name	Form Type	Date Filed	File Name
-----	--------------	-----------	------------	-----------

SAMPLE ROWS

1000209	MEDALLION FINANCIAL CORP	8-K	20260626	edgar/data/1000209/0001193125-26-284003.txt
1122020	PCS Edventures!, Inc.	10-K	20260626	edgar/data/1122020/0001493152-26-030242.txt
1022837	SUMITOMO MITSUI FINANCIAL GROUP, INC.	20-F	20260626	edgar/data/1022837/0001193125-26-283871.txt
1000275	ROYAL BANK OF CANADA	424B2	20260626	edgar/data/1000275/0000950103-26-009532.txt

✓ **These five fields tell us**

- "A given form appeared under a given CIK today, and the complete file path is here."

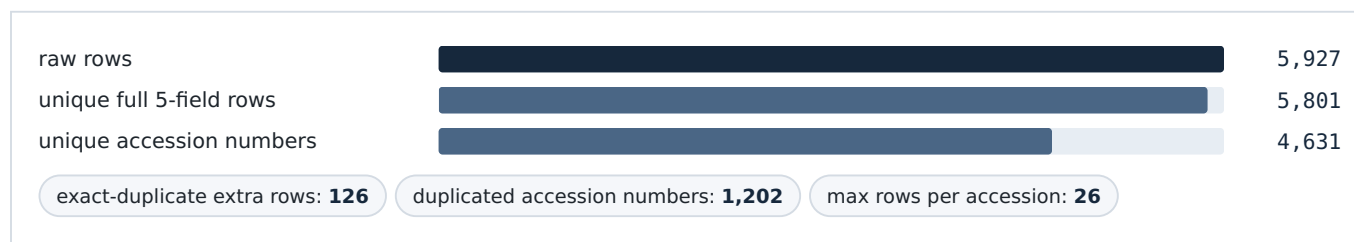
✗ **They do not tell us**

- how many documents the filing contains
- who the actual issuer is
- whether there is a reporting owner
- which item(s) an 8-K reports
- which file is the primary document
- which file is the XBRL instance

All of that requires continuing to parse down the accession materials.

4 Raw Daily Index Rows Must Be Staged First

The fixed sample for 2026-06-26 shows:



This shows that a raw daily-index row is not a clean business event. Exact duplicate rows can occur, and the same accession can hang under multiple CIKs. Common scenarios for cross-CIK joint filings include:

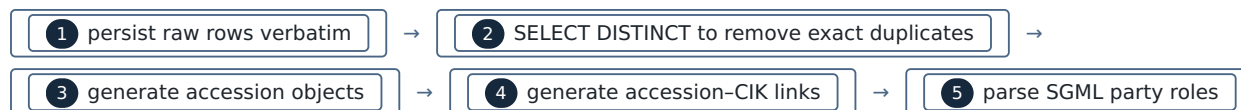
- **Form 3/4/5:** a single ownership filing by an executive, director, or major shareholder involves both the reporting owner and the issuer.
- **SC 13D / SC 13G:** a single large-stake beneficial-ownership disclosure involves the filed-by / reporting person and the subject company.
- **Fund or fund-series filings:** one submission may correspond to multiple funds, series, or classes.
- **Joint or group filings:** multiple entities filing together.
- **M&A, tender-offer, and subject-company filings:** the submitting party and the affected party may differ.

Therefore, the system's first step must be to persist raw rows verbatim, and only after that generate normalized rows:

raw_daily_index_row:	normalized_daily_row:
source_date	source_date
line_number	row_cik
raw_line	row_company_name

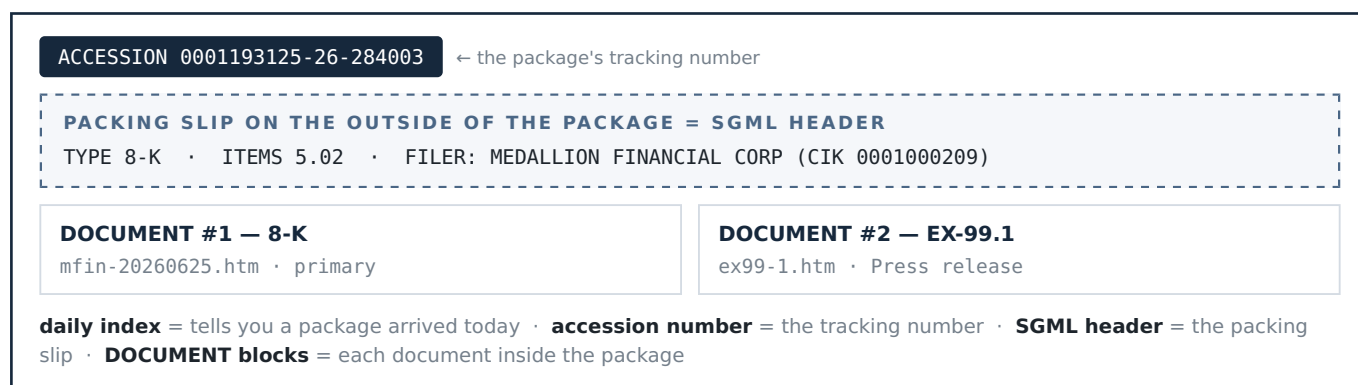
raw_line_hash	form_type
cik_raw	filed_date
company_name_raw	file_name
form_type_raw	accession_number
filed_date_raw	accession_no_dash
file_name_raw	complete_text_url
ingest_timestamp	hdr_sgml_url
	directory_index_json_url

The key rule: **do not use accession_number alone as the raw-row primary key, and do not treat the daily-row CIK as the business issuer.** The correct sequence is:



5 What SGML Actually Is

SGML stands for Standard Generalized Markup Language — think of it as a predecessor-generation markup language to HTML and XML. Inside EDGAR, it is not the financial-report body meant for human reading; it is the structured envelope SEC uses to package, transmit, and distribute a filing. A simple mental model — a filing is a package:



A submission looks roughly like this:

```

<SUBMISSION>
  <ACCESSION-NUMBER>0001193125-26-284003
  <TYPE>8-K
  <PUBLIC-DOCUMENT-COUNT>...
  <ITEMS>5.02

  <FILER>
    <COMPANY-DATA>
      <CONFORMED-NAME>MEDALLION FINANCIAL CORP
      <CIK>0001000209
      <ASSIGNED-SIC>...
      <FISCAL-YEAR-END>...

  <DOCUMENT>
    <TYPE>8-K
    <SEQUENCE>1
    <FILENAME>mfin-20260625.htm
    <DESCRIPTION>8-K
  
```

```

</DOCUMENT>

<DOCUMENT>
  <TYPE>EX-99.1
  <SEQUENCE>2
  <FILENAME>ex99-1.htm
  <DESCRIPTION>Press release
</DOCUMENT>
</SUBMISSION>

```

The header of the actual complete `.txt` is sometimes in a different, colon-delimited format:

```

ACCESSION NUMBER: 0001104659-20-000849
CONFORMED SUBMISSION TYPE: 8-K
PUBLIC DOCUMENT COUNT: 41
ITEM INFORMATION: Regulation FD Disclosure
ITEM INFORMATION: Financial Statements and Exhibits
FILED AS OF DATE: 20200103
FILER:
  COMPANY DATA:
    COMPANY CONFORMED NAME: Helmerich & Payne, Inc.
    CENTRAL INDEX KEY: 0000046765

```

So the SGML parser must not assume a single format. The `.hdr.sgml` may look more like XML/SGML tags, while the header of the complete `.txt` may look more like colon-delimited fields. They are two surface forms of the same class of metadata.

6 How to Build the hdr.sgml Parser

The goal of the **hdr.sgml parser** is not to parse the financial-report body but to parse the submission's "packing slip." It must answer: what is this filing, who submitted it, whom is it about, which files does it contain, and which items does the 8-K report.

Recommended implementation: rule-based parsing first — no AI.

1. Prefer requesting the `.hdr.sgml` under the accession directory.
2. If `.hdr.sgml` does not exist, request the first 64KB / 256KB of the complete `.txt` as a capped read.
3. Parse submission-level fields first: accession, type, filed date, acceptance datetime, period, document count.
4. Then parse party blocks: FILER, ISSUER, REPORTING OWNER, SUBJECT COMPANY, FILED BY.
5. Then parse DOCUMENT blocks: TYPE, SEQUENCE, FILENAME, DESCRIPTION.
6. If the form is an 8-K, parse ITEMS or ITEM INFORMATION.
7. Store source and confidence for every field.

The parser must be lenient — do not force a strict XML parser onto this input. Historical EDGAR text can mix legacy formats, colon-delimited fields, repeated blocks, casing differences, and non-strict XML structure. The output should flow into these structured tables:

filing_accession	accession_party	filing_document	eight_k_item
------------------	-----------------	-----------------	--------------

7 What the accession_party and filing_document Tables Are

accession_party is the table for "which parties this filing involves, and what role each party plays." It solves the entity-identification problem.

```
accession_party:
  accession_number
  party_role = FILER / ISSUER / REPORTING_OWNER / SUBJECT_COMPANY / FILED_BY
  cik
  conformed_name
  sic
  state_of_incorporation
  fiscal_year_end
  source
  parse_confidence
```

Its value is most visible in Form 4, 13D/13G, and multi-party filings. In a Form 4:

REPORTING_OWNER

the executive, director, or major shareholder

files an ownership
report about →

ISSUER

the securities issuer, i.e. the company being
monitored

WHY THIS TABLE EXISTS

Without `accession_party`, the system could mistake an individual executive for the insured company, or treat a 13D's filed-by party as the subject company.

filing_document is the table for "which files this filing contains." It solves the material-enumeration and parser-dispatch problem.

```
filing_document:
  accession_number
  sequence
  document_name
  document_type_from_sgml
  description_from_sgml
  document_class
  is_primary_document
  is_xbrl_instance
  is_inline_xbrl_html
  is_ownership_xml
  document_url
  download_status
  parse_status
  parse_error
```

A single accession may contain the primary HTML, exhibits, the XBRL instance, schema, label, calculation, presentation, FilingSummary.xml, PDFs, images, and more. `filing_document` lets the system know which parser each file should be routed to, which files should be deferred, and which are the critical materials.

8 What the Primary Document Is, and Why a Detector Must Be Built

The **primary document** is the main file of the filing. For a 10-K it is usually the annual-report body; for an 8-K it is the Form 8-K body; for a 20-F it is the annual-report body. It is not an exhibit, not a signature attachment, not an XBRL schema, and not a label/calculation XML.

Why must a detector be built? Because an accession directory can contain dozens of files, and file names are not always reliable. In testing we found that relying on "the first non-exhibit HTML" alone can wrongly grab an exhibit such as `ex31-1.htm` instead of the main document. The `type` field in `index.json` may also be nothing more than an icon name, e.g. `text.gif`, and cannot be treated as the SEC document type.

For the MVP a rule-based detector is the better choice; do not start with AI. Priority order:

- 1

SGML `<DOCUMENT><TYPE>` matches the form type

structural · high
- 2

The Document Type on the filing detail page matches the form type

structural · high
- 3

FilingSummary.xml points to the report instance

structural
- 4

Filename / description heuristics

low confidence only

Every detector run outputs:

primary_document_name	primary_detection_method	primary_detection_confidence	reason
-----------------------	--------------------------	------------------------------	--------

AI can be used later for comparison testing, especially on low-confidence samples, old historical files, and unusual document types. But the demo should not rely on AI for primary-document detection, because this step needs to be deterministic, auditable, and reproducible. Later, a golden set can be built to compare the rule-based detector against an AI detector on precision/recall, and only then decide whether to introduce AI as a fallback.

9 Why the 8-K Item Extractor Is a Separate Component

The 8-K is the current-report form. For insurance and underwriting risk, the events commonly disclosed in 8-Ks can surface risk earlier than financial-statement numbers do: bankruptcy, debt default, auditor changes, restatements, delisting, executive departures, material agreements, major financings, and so on.

The 8-K item does not appear in the daily master index. It typically lives in three places:

1. The numeric tag in `.hdr.sgml`, e.g. `<ITEMS>5.02`.
2. The descriptive text in the complete `.txt` header, e.g. `ITEM INFORMATION: Departure of Directors or Certain Officers...`
3. The headings inside the primary-document body, e.g. Item 5.02 / Item 4.02.

The measured sample shows, for the same MFIN 8-K:

<code>.hdr.sgml:</code>	<code><ITEMS>5.02</code>
<code>complete .txt:</code>	<code>ITEM INFORMATION: Departure of Directors or Certain Officers...</code>

So the extractor must support three-tier fallback:

- 1

numeric codes from `.hdr.sgml <ITEMS>`

high confidence
- 2

ITEM INFORMATION descriptive text from the complete `.txt` — descriptor → code mapping

mapped
- 3

primary-document heading regex

medium / low

Output table:

```
eight_k_item:
  accession_number
  item_code
  item_text_raw
  extraction_source = hdr.sgml / txt_header / primary_heading
  mapping_method = direct_numeric / descriptor_mapping / heading_regex
  confidence
  fallback_used
```

Worked example — one item becomes one risk signal

0001193125-26-284003 · 8-K · source: `hdr.sgml` → `item_code 5.02` → Departure of Directors or Certain Officers

→ `risk_signal = management_change`

multi-item example: 1.01 Material Definitive Agreement + 2.03 Direct Financial Obligation + 9.01 Exhibits

The system must allow a single 8-K to carry multiple items — do not store only one `item_code`.

10

Designing the XBRL / iXBRL Instance Detector

XBRL is the XML format for financial facts; **iXBRL** is inline XBRL, i.e. the tags embedded directly inside the HTML report. Per SEC's documentation, XBRL may exist as a standalone XML file or be embedded in quarterly and annual HTML reports; SEC's XBRL APIs aggregate facts that use non-custom taxonomies and apply to the entire filing entity.

An accession directory can contain many XML files, but not every XML is a financial instance:

File in the accession directory	What it is	Feed to fact extractor?
form10-k.htm	primary HTML / inline XBRL HTML	× — HTML carrier
form10-k_htm.xml	extracted inline XBRL instance	✓ the instance
company-20260331.xsd	taxonomy schema	×
company-20260331_cal.xml	calculation linkbase	×
company-20260331_lab.xml	label linkbase	×
company-20260331_pre.xml	presentation linkbase	×
FilingSummary.xml	XBRL report summary	×
ownership.xml	Form 4 ownership XML — not financial XBRL	×

Without an instance detector, the fact extractor can easily grab the wrong file — e.g. treating a schema, label, calculation, or ownership XML as the financial instance. The detector's responsibilities:

1. Decide whether the accession should have financial XBRL at all.
2. Locate the true instance document.
3. Distinguish standalone XBRL from extracted inline-XBRL instances.
4. Hand the fact extractor an unambiguous input.
5. For financial-report filings with no instance, record `XBRL_INSTANCE_NOT_FOUND`.

11 How to Design the XBRL Fact Extractor

The first version of the XBRL fact extractor does not need a full financial model, but it must prove that "real facts can be extracted correctly." Minimal design:

```
input: xbrl_instance.xml or an extracted inline XBRL instance

parse:
  contexts: contextRef -> period / instant / dimensions
  units: unitRef -> unit
  facts: concept / namespace / contextRef / unitRef / decimals / value

normalize:
  taxonomy = us-gaap / ifrs-full / dei / custom
  period_start / period_end / instant
  value_numeric

classify candidates:
  cash · assets · liabilities · revenue · net_income · equity

output:
  xbrl_instance_fact
  xbrl_fact_extraction_summary
```

Verified example — PCS Edventures 10-K

CIK 1122020 · accession 0001493152-26-030242 · primary document form10-k.htm · XBRL instance form10-k_htm.xml

673 total facts parsed from one instance



LOCKED BY GOLDEN ASSERTION (\$16)

Cash @ 2026-03-31 = 2,674,538

This shows that the fact extractor cannot stop at "an XBRL file was found to exist." It must parse contexts, units, concepts, and values, and lock at least a few fixed values with golden assertions.

12 isXBRL=true Does Not Mean companyfacts Already Has Usable Facts

This trap was discovered through TSM's FPI/IFRS case.



The causes fall into several classes:

1. `isXBRL` is a filing-level flag meaning the filing carries XBRL-related material; it does not mean companyfacts has already aggregated usable company-level financial facts.
2. SEC companyfacts aggregates only facts that use non-custom taxonomies and apply to the entire filing entity; if the material is mainly custom taxonomy, dimensional facts, exhibit-attached facts, filing fees, cover pages, or non-financial XML, the target facts may simply be absent.
3. The FPI/IFRS mandatory timeline arrived late. The IFRS taxonomy entered mandatory structured coverage only for fiscal years ending on or after 2017-12-15, so companies like TSM show an inconsistency between the submissions flag and companyfacts availability.
4. companyfacts also has processing latency. SEC documentation states the API is generally updated in real time but may be slower at peak times; the bulk ZIP is recompiled nightly.
5. Some filings carry XBRL/iXBRL that is simply not the financial facts we need — e.g. 8-K cover pages, filing-fee extracts, ownership XML, or employee-benefit-plan filings.

✗ Do not write

```
if submissions.isXBRL == true:
    data_available = true
```

✓ Write instead

```
if isXBRL:
    detect actual instance
    parse facts
    classify useful facts
    crosscheck companyfacts by accession
    if no useful facts:
        status =
        XBRL_EXISTS_BUT_NO_USEFUL_FACTS
```

13 What FPI / IFRS Issuers Are

An **FPI** is a foreign private issuer — put simply, a foreign company that discloses or lists in U.S. capital markets but is not a U.S. domestic issuer. FPIs generally do not file 10-K / 10-Q; instead they file:

```
20-F: annual report of foreign issuers
40-F: annual report of Canadian issuers under MJDS
      (the U.S.-Canada Multijurisdictional Disclosure System)
6-K:  interim information furnished to SEC by foreign issuers
```

IFRS stands for International Financial Reporting Standards. Many foreign issuers report under IFRS, but not every FPI does. In our tests, ASML is a foreign company yet its companyfacts taxonomy is us-gaap; TSM, by contrast, exhibits the FPI/IFRS gap between the 2016 flag and the 2018 companyfacts availability.

Issuer type	Typical filings	Structured (XBRL) coverage
U.S. domestic operating company	10-K + 10-Q	quarterly cadence is fairly stable
FPI	20-F / 40-F + 6-K	XBRL financial data skews annual; quarterly structured coverage is weak
IFRS FPI	20-F + 6-K	the pre-2018 structured-coverage gap is even larger

IMPLICATION FOR THE PRICING MODEL

The pricing model must not apply the same quarterly data-freshness assumptions to FPIs that it applies to U.S. companies.

14 Is companyfacts an API or a URL? Does It Return All Years?

SEC officially refers to the submissions and extracted XBRL data on `data.sec.gov` as RESTful APIs. The concrete entry point of `companyfacts` is a URL:

```
https://data.sec.gov/api/xbrl/companyfacts/CIK#####.json
```

There is no contradiction between "URL" and "API" here. The URL is the access address; the API is the contract behind that address: input a CIK, get back JSON whose structure contains that company's company concepts and facts within the SEC XBRL dataset.

SEC's documentation states that the companyfacts API "returns all the company concepts data for a company into a single API call". That "all" must be read within the boundaries of the SEC XBRL API:

✓ Included

- the company concepts / facts available for that CIK within the SEC XBRL API dataset, typically spanning multiple years and multiple accessions

✗ Not included

- most pre-2009 history
- unstructured disclosures
- the full contents of custom taxonomies
- non-entire-entity facts
- all HTML tables, PDFs, exhibits, and narrative text

So it is fair to say: companyfacts is SEC's company-level XBRL facts API endpoint, and it typically returns the standard company concepts across all available years for that company. It is not fair to say it returns "all financial information in the company's history."

15 Why the companyfacts Crosscheck Exists, and How to Handle Non-Reconcilable Cases

companyfacts contains facts for historical periods, but the payload itself is SEC's current aggregation as handed to you today. It does not tell you on which historical date a given fact first became visible, when it was corrected, or when SEC re-extracted it. Reading the latest companyfacts directly therefore introduces look-ahead bias.

The companyfacts crosscheck is thus not a substitute for local accession parsing; it exists for three things:

1. Check whether the local XBRL parser extracted correctly.
2. Determine whether SEC's aggregation layer already carries facts for the same accession.

3. Identify coverage gaps and point-in-time risk.

The handling logic, as a decision matrix over the two evidence sources:

	companyfacts HAS the same accession	companyfacts LACKS the same accession
local parser HAS facts	A — values consistent → compare concept / unit / period / value; if consistent, high confidence . D — values disagree → check unit, period, context, dimension, decimals, scale. Do not auto-overwrite; preserve the mismatch evidence.	B → mark <code>COMPANYFACTS_GAP</code> . Do not discard the local facts. Retry after 24 hours or after the next nightly bulk. If still absent, classify the cause: custom taxonomy · non-entire-entity facts · FPI/IFRS gap · a filing type that never enters companyfacts · an SEC processing gap.
local parser LACKS facts	C → this is local parser risk; mark <code>PARSER_GAP</code> . Route the accession to manual/engineering review or the parser-fix queue.	— not an enumerated case

STATUS E — ORTHOGONAL TO THE MATRIX

`XBRL_EXISTS_BUT_NO_USEFUL_FACTS` — the instance was detected and parsed, but zero useful candidates were classified. Determine whether it is a filing fee, a cover page, an 11-K, narrow 8-K tagging, custom-only taxonomy, or a non-target financial statement.

In the production system, the companyfacts crosscheck status should be written into `xbml_fact_extraction_summary`, not merely into logs.

16

Why Golden Assertions Are Necessary

Golden assertions lock verified ground-truth values into automated tests. They protect against the most dangerous class of failure: the system looks like it runs, but the numbers are wrong. Without golden assertions, none of the following would raise an error:

- ✗ The daily-index parser reads one column short but still emits a CSV.
 - ✗ Accession deduplication uses the wrong key and swallows multi-CIK relationships.
 - ✗ The primary detector picks an exhibit as the main document.
 - ✗ The 8-K item extractor keeps only the first item and misses multi-item filings.
 - ✗ The XBRL parser extracts 100 facts but silently misses another 500.
 - ✗ The companyfacts crosscheck checks that a cash field exists but never checks whether the value is correct.

The MVP should lock at least the following:

- ☐ 2026-06-26 daily index: raw rows = 5927, unique accessions = 4631, max rows per accession = 26
 - ☐ Apple first XBRL 10-Q = 2009-07-22; first XBRL 10-K = 2009-10-27
 - ☐ Apple Form 3/4/SD isXBRL count = 0
 - ☐ PCS 10-K accession 0001493152-26-030242: fact_count = 673, Cash = 2674538
 - ☐ TSM: the gap between submissions isXBRL (2016) and earliest companyfacts facts (2018) holds
 - ☐ A fixed 8-K accession: item extraction returns the expected item codes

Golden assertions do not replace monitoring, but they prevent the ingestion MVP from silently regressing as it iterates.

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What the Saturday Full Index Validates Against the Daily Index

Per SEC's official documentation: a filing may be removed or corrected — at the filer's request and with SEC staff authorization — because it was submitted under the wrong filer, duplicated, unreadable, contains sensitive information, or for other reasons. Corrections processed within the same business day enter that evening's index. If a removal occurs on a later business day, it is not reflected in prior daily, feed, or oldload indexes. The full and quarterly index files are rebuilt every Saturday morning to incorporate the deletes and updates from post-acceptance corrections (PAC).

MON pull daily master index → raw staging	TUE pull daily master index	WED pull daily master index	THU pull daily master index	FRI pull daily master index	SAT SEC rebuilds full + quarterly indexes (PAC deletes / updates land here)	SUN —
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→ after each Saturday rebuild: pull the current-quarter full index and reconcile it against locally accumulated daily rows

Therefore, the Saturday full-index reconciliation primarily validates the discrepancies created by SEC's after-the-fact processing — it does not validate all errors in our own parser. It helps the system discover:

1. An accession that later disappears from the full index — possibly a PAC delete.
2. A filing whose metadata has changed.
3. An accession that appeared in the daily index but is no longer in the current-quarter full index.
4. Filings that the weekday daily job missed.

DISCREPANCY HANDLING RULE

Never hard-delete; mark removed_by_sec / corrected / metadata_changed / missed_by_daily_job.

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Recommended SEC Data Acquisition When Direct Web Scraping Is Not Permitted

If corporate policy does not allow business teams to scrape data directly from the web, do not let application services connect to SEC directly. The recommendation is for the data team to provide a compliant wrapper or data product.

SEC official endpoints / Archives**Data Team SEC Connector / Wrapper**

corporate User-Agent + contact email · fair-access rate limiting · caching · retries · ETag/Last-Modified · hashing · request logging

**Raw landing zone**

daily index · hdr.sgml · index.json · companyfacts · submissions · selected documents

**Curated tables**

normalized_daily_row · filing_accession · accession_party · filing_document · xbrl_fact

**Application / risk model**

reads the internal API or warehouse tables

The data-team wrapper should own:

1. A unified corporate User-Agent and contact email.
2. SEC fair-access rate limiting: no more than 10 requests per second; 5 requests per second recommended.
3. Caching, retries, ETag/Last-Modified, hashing, request logging.
4. Landing raw files and preserving provenance.

19 Design Recommendations

Build the ingestion loop first — the numbering below is the build order:

1 daily index raw staging

2 normalized accession + CIK link

3 hdr.sgml parser

4 accession_party table

5 filing_document table

6 primary document detector

7 8-K item extractor

8 XBRL / iXBRL instance detector

9 XBRL fact extractor

10 companyfacts crosscheck

11 golden assertions

12 weekly full-index reconciliation

Component	Layer it plays in the system
daily index	the discovery layer
SGML header	the "who + what happened" metadata layer
filing_document	the materials layer
primary document	the entry point into the report body
8-K items	the event-classification layer
XBRL instance	the accession-level source of financial facts
companyfacts	SEC's company-level aggregation-and-reconciliation layer
weekly full index	the PAC correction layer

The route is viable — but the MVP must be built by decomposing it along exactly these layers.